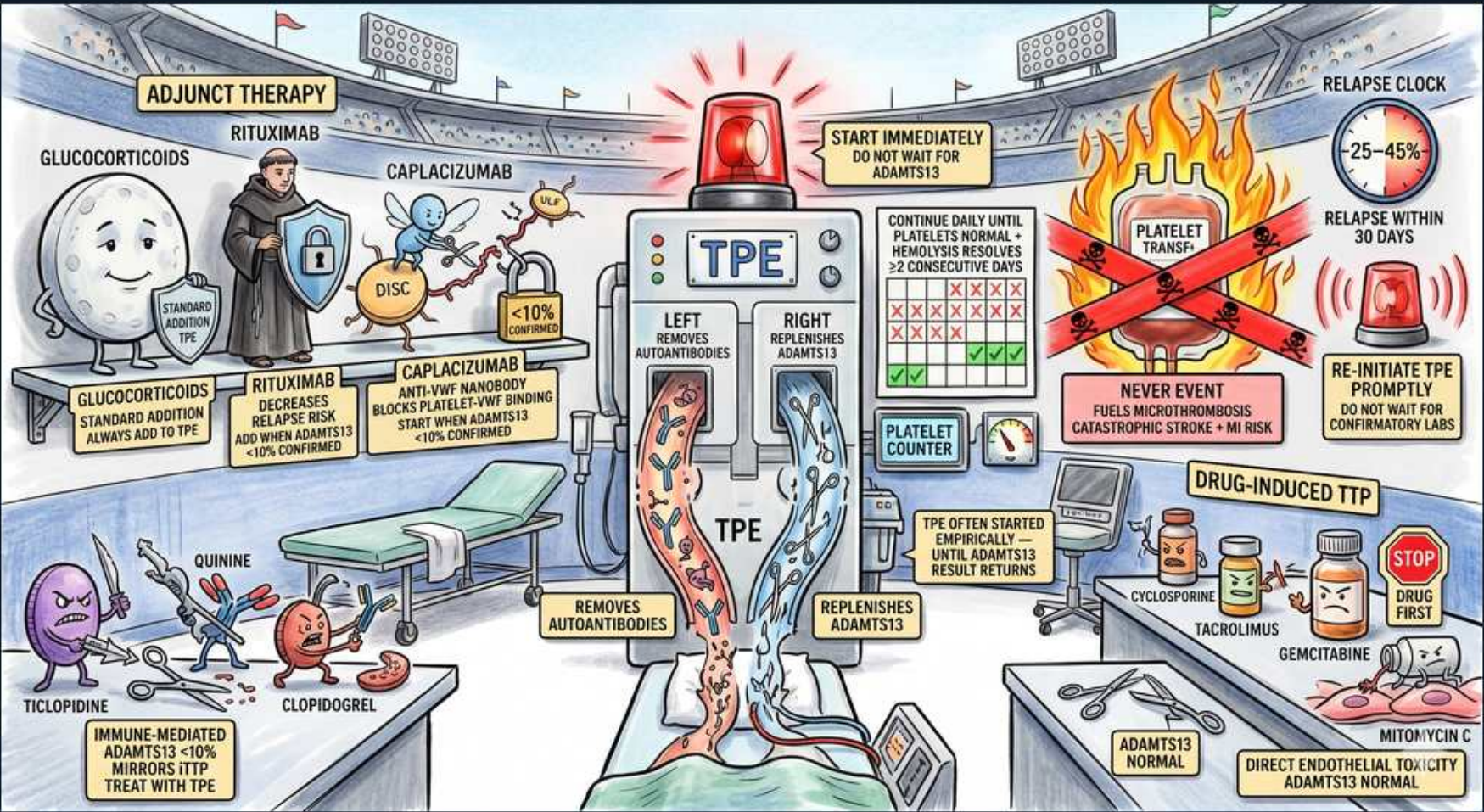


Platelet disorders — TTP treatment (TPE)



1. Glucocorticoids

WHAT YOU SEE

A crescent-moon GLUCOCORTICOID pill in the ADJUNCT THERAPY shelf — the moon-pill steroid added on top of plasma exchange in essentially all acquired TTP.

WHAT IT ENCODES

Corticosteroids (e.g., high-dose methylprednisolone or prednisone ~1 mg/kg) are a STANDARD add-on to plasma exchange in essentially all confirmed acquired TTP. Rationale: TTP is autoantibody-driven, so steroids suppress production of the anti-ADAMTS13 inhibitor while PLEX removes existing antibody and replaces enzyme.

BOARD TRAP

Treating TTP with steroids ALONE (or PLEX alone) is the trap — neither single modality is adequate for acute acquired TTP. Steroids address the autoantibody but do not acutely clear UL-vWF multimers or replace ADAMTS13; they are an adjunct TO plasma exchange, never a substitute for it.

2. Rituximab

WHAT YOU SEE

A hooded MONK labeled RITUXIMAB captioned 'DECREASES RELAPSE RISK' — the anti-CD20 B-cell killer that lowers recurrence.

WHAT IT ENCODES

Rituximab (anti-CD20) depletes the B cells making anti-ADAMTS13 antibodies. It is added in confirmed acquired TTP — especially relapsing, refractory, or severe disease — and reduces RELAPSE and shortens time to remission. Increasingly used early in immune TTP.

BOARD TRAP

Expecting rituximab to act FAST in the acute crisis is the trap — its B-cell-depleting effect builds over days to weeks, so it does not replace the immediate, life-saving roles of plasma exchange and steroids. It modifies the relapse trajectory; it is not the acute rescue. Screen hepatitis B before giving it.

3. Caplacizumab

WHAT YOU SEE

A padLOCK/nanobody figure labeled CAPLACIZUMAB 'BLOCKS PLATELET-vWF BINDING' — the lock clamps vWF so platelets can't stick.

WHAT IT ENCODES

Caplacizumab is an anti-vWF NANOBODY that blocks the A1 domain of von Willebrand factor, preventing platelet binding to UL-vWF multimers. Added to PLEX + immunosuppression, it speeds platelet recovery and reduces TTP-related death, recurrence, and refractoriness. Its main adverse effect is mucocutaneous BLEEDING (it impairs platelet-vWF adhesion).

BOARD TRAP

Forgetting that caplacizumab causes bleeding (it blocks platelet-vWF binding) and continuing it through a procedure or active hemorrhage is the trap. Also note caplacizumab treats the downstream platelet clumping but does NOT correct the underlying ADAMTS13 deficiency — it is layered on top of PLEX and immunosuppression, not a stand-alone cure.

4. TPE start immediately

WHAT YOU SEE

The big central TPE / plasma-exchange machine under a red alarm light, stamped 'START IMMEDIATELY / DO NOT WAIT FOR ADAMTS13' — the life-saving exchanger fired up at once.

WHAT IT ENCODES

Plasma exchange (PLEX / TPE / plasmapheresis) is THE life-saving therapy and must be STARTED IMMEDIATELY on clinical suspicion (MAHA + thrombocytopenia). It does two things at once: removes the anti-ADAMTS13 autoantibody and the UL-vWF multimers, and replaces functional ADAMTS13 via donor plasma. Untreated mortality ~90% drops to ~10–20% with prompt PLEX.

BOARD TRAP

Delaying PLEX to wait for the ADAMTS13 assay (or for more pentad features) is the single most-tested fatal error. Draw ADAMTS13 first, then start empiric exchange the same day. If PLEX is not instantly available, give plasma INFUSION as a bridge — but never simply wait. 'Hold therapy pending confirmation' is always wrong.

5. TPE removes antibody

WHAT YOU SEE

The LEFT side of the TPE machine labeled 'REMOVES AUTOANTIBODIES' — the outflow that strips the patient's anti-ADAMTS13 antibody and UL-vWF.

WHAT IT ENCODES

One half of why PLEX works: it physically REMOVES the patient's plasma containing anti-ADAMTS13 autoantibodies (and the pathogenic UL-vWF multimers), discarding the drivers of microthrombosis. This clearance is something simple plasma INFUSION cannot achieve.

BOARD TRAP

Choosing plasma INFUSION instead of plasma EXCHANGE in acquired (autoantibody) TTP is the trap — infusion supplies ADAMTS13 but cannot remove the inhibitor or the UL-vWF load, so it is inferior and used only as a temporary bridge when exchange is unavailable. Exchange is required because removal of antibody is half the mechanism.

6. TPE replenishes ADAMTS13

WHAT YOU SEE

The RIGHT side of the TPE machine labeled 'REPLENISHES ADAMTS13' — the donor-plasma inflow that restores the missing enzyme.

WHAT IT ENCODES

The other half: the donor plasma used as replacement fluid REPLENISHES functional ADAMTS13, restoring the enzyme that cleaves UL-vWF. Removal of inhibitor + repletion of enzyme together break the microthrombotic cycle. This is also why congenital TTP (no antibody) can be managed with plasma INFUSION alone — they just need the enzyme.

BOARD TRAP

Conflating the two TTP forms is the trap: acquired (antibody) TTP needs EXCHANGE (remove antibody + replace enzyme), whereas congenital Upshaw-Schulman TTP, which has no inhibitor, can often be treated with simple plasma INFUSION or recombinant ADAMTS13 to supply the missing enzyme. Prescribing infusion-only for acquired TTP under-treats it.

7. Continue until resolved

WHAT YOU SEE

A row of check-marks captioned 'CONTINUE DAILY UNTIL PLATELETS NORMAL / HEMOLYSIS RESOLVES x2 CONSECUTIVE DAYS' — keep exchanging until counts hold normal.

WHAT IT ENCODES

Continue DAILY plasma exchange until the platelet count NORMALIZES (commonly >150,000 for ≥2 consecutive days) and hemolysis resolves (LDH normalizing). Track response with platelet count and LDH. Stopping too early invites exacerbation; taper/discontinue only after sustained recovery.

BOARD TRAP

Stopping PLEX as soon as the first platelet count ticks up is the trap — premature discontinuation risks early exacerbation. The endpoint is sustained normalization (platelets normal ×2 days plus resolving hemolysis/LDH), not a single improved value. Rising LDH or falling platelets after stopping signal you stopped too soon.

8. NO platelet transfusion

WHAT YOU SEE

A bag of platelets engulfed in FLAMES and crossed out, labeled 'PLATELET TRANS / NEVER EVENT / FUELS CATASTROPHIC STROKE, MI' — transfusing platelets pours fuel on the clotting fire.

WHAT IT ENCODES

Platelet transfusion is CONTRAINDICATED in TTP — adding platelets pours fuel on the fire, feeding the vWF-driven microthrombi and precipitating catastrophic arterial thrombosis (STROKE, MI, worsening organ ischemia). This is regarded as a 'never event' absent truly life-threatening hemorrhage.

BOARD TRAP

The board-favorite trap: a TTP patient with profound thrombocytopenia, so the tempting (wrong) answer is to transfuse platelets to 'correct' the number. Do NOT — it can cause fatal thrombosis. Treat with PLEX. Contrast with ITP, where platelets are merely ineffective but permitted in emergencies; in TTP they are actively harmful.

9. Relapse window

WHAT YOU SEE

A RELAPSE CLOCK marked '25-45%' and 'WITHIN 30 DAYS' — the alarm clock showing how often and how soon TTP comes back.

WHAT IT ENCODES

TTP relapses are common — roughly 25–45% recur, frequently within the first ~30 days (exacerbation) and sometimes months to years later. Persistently low ADAMTS13 activity in remission predicts relapse, so patients are monitored with serial counts/LDH and sometimes ADAMTS13 levels.

BOARD TRAP

Assuming a patient is 'cured' after the first remission and skipping follow-up is the trap. The high relapse rate (25–45%), especially in the first month, mandates ongoing surveillance; rituximab (and pre-emptive treatment of falling ADAMTS13) is used specifically to lower this relapse risk. Discharging without a monitoring plan is wrong.

10. Re-initiate TPE on relapse

WHAT YOU SEE

A 'RE-INITIATE TPE PROMPTLY / DO NOT WAIT FOR CONFIRMATORY LABS' callout — restart the exchanger on clinical relapse without awaiting ADAMTS13.

WHAT IT ENCODES

On relapse or exacerbation, RE-INITIATE plasma exchange PROMPTLY on clinical grounds (returning MAHA + thrombocytopenia) — do not wait for confirmatory ADAMTS13 results, exactly as in the first presentation. Restart/intensify immunosuppression (steroids, rituximab) as well.

BOARD TRAP

Waiting for repeat ADAMTS13 confirmation before restarting PLEX at relapse repeats the original cardinal error — the same urgency applies. Falling platelets and rising LDH in a known TTP patient are enough to restart exchange immediately; 'await labs' is wrong both at first presentation and at relapse.

11. Quinine / ticlopidine cause

WHAT YOU SEE

QUININE, TICLOPIDINE, and CLOPIDOGREL bottles labeled 'IMMUNE-MEDIATED / ADAMTS13 <10% / MIRRORS ITTP / TREAT WITH TPE' — the immune drug-TTP that needs exchange just like primary TTP.

WHAT IT ENCODES

IMMUNE-mediated drug-induced TTP — classically QUININE, and the thienopyridines TICLOPIDINE and clopidogrel — produces antibodies against ADAMTS13, so ADAMTS13 is <10% and the picture MIRRORS idiopathic TTP. Management is the same: STOP the drug AND treat with plasma exchange (± steroids/rituximab).

BOARD TRAP

The trap is to merely stop the drug and observe in immune-type drug-induced TTP — these (quinine, ticlopidine, clopidogrel) have ADAMTS13 <10% and need PLEX just like primary TTP. Failing to start exchange because 'it's just a drug reaction' is wrong. Distinguish this immune mechanism from the direct-toxicity drugs at the other end.

12. Direct toxicity drugs

WHAT YOU SEE

GEMCITABINE, CYCLOSPORINE/TACROLIMUS, and MITOMYCIN C bottles labeled 'DIRECT ENDOTHELIAL TOXICITY / ADAMTS13 NORMAL / STOP DRUG FIRST' — direct-toxic TMA where you pull the drug, not exchange.

WHAT IT ENCODES

DIRECT endothelial-toxic drugs — gemcitabine, mitomycin C, calcineurin inhibitors (cyclosporine/tacrolimus), and others — cause a thrombotic microangiopathy WITHOUT antibody and with NORMAL ADAMTS13 (>10%). Because there is no inhibitor and no enzyme deficiency, the cornerstone is to STOP the offending drug; plasma exchange is much LESS effective here.

BOARD TRAP

Reflexively starting plasma exchange for gemcitabine/mitomycin/calcineurin-inhibitor TMA is the trap — ADAMTS13 is NORMAL, the mechanism is direct endothelial injury, and PLEX is of limited benefit. The key intervention is DRUG WITHDRAWAL (and supportive care). The discriminator from the immune drug group is the ADAMTS13 level: <10% immune (PLEX) vs normal direct-toxic (stop drug).